

CARIBOU ISLAND, ON, Canada

WMO: 714330

Lat: 47.339N

Lon: 85.826W

Elev: 189

StdP: 99.08

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.4	-19.9	0.7	-17.3	-17.2	0.8	-14.5	18.4	-4.3	17.0	-3.5	6.6	10	0.831

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	21.2	18.5	20.0	17.9	18.9	17.1	19.3	20.5	18.2	19.6	17.4	18.7	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.9	20.0	17.7	13.0	19.0	16.7	12.2	18.2	55.8	20.5	52.4	19.6	49.7	18.8	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.1	12.6	-19.6	23.5	2.4	1.6	-21.3	24.7	-22.7	25.7	-24.0	26.6	-25.7	27.8		
(5)			-19.8	20.2	2.4	1.5	-21.6	21.2	-23.0	22.1	-24.3	22.9	-26.1	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-5.7	-5.3	-2.3	2.1	6.0	9.1	12.9	15.9	13.6	7.8	2.4	-2.0
	DBStd	8.06	4.95	5.06	4.70	2.88	2.17	2.38	3.52	2.84	2.92	3.25	3.85	4.36
	HDD10.0	2408	486	429	382	236	127	45	11	1	5	84	229	373
	HDD18.3	5030	744	663	640	486	384	277	172	84	142	327	479	632
	CDD10.0	433	0	0	0	0	1	19	101	183	114	16	0	0
	CDD18.3	13	0	0	0	0	0	0	3	8	1	0	0	0
	CDH23.3	3	0	0	0	0	0	0	1	2	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.4	8.1	7.9	6.5	6.2	5.5	4.8	4.5	4.9	5.9	7.1	7.5	8.0
Precipitation	PrecAvg	1076	91	68	73	90	91	91	86	80	105	115	93	94
	PrecMax	1364	161	127	176	151	140	189	156	129	224	185	149	175
	PrecMin	841	41	19	12	25	38	43	34	31	44	53	41	15
	PrecStd	126	29	27	36	36	28	31	31	25	35	40	26	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.0	7.8	10.5	13.8	17.7	22.5	23.4	20.9	16.1	10.6	6.2
		MCWB	3.5	3.3	6.3	6.5	10.6	14.2	19.0	19.9	18.2	14.7	9.6	5.4
	2%	DB	3.1	3.0	5.5	7.9	11.7	15.6	20.7	21.7	19.4	14.7	9.2	5.2
		MCWB	2.5	2.4	4.3	5.5	9.2	13.0	18.0	19.1	17.4	13.3	8.4	4.4
	5%	DB	2.2	2.2	4.1	6.6	10.3	14.1	19.3	20.6	18.4	13.2	8.1	4.5
		MCWB	1.6	1.7	3.0	4.7	8.4	12.1	17.1	18.4	16.8	12.0	7.2	3.8
	10%	DB	1.1	1.2	3.0	5.5	9.1	12.8	17.9	19.5	17.5	12.2	7.3	3.7
		MCWB	0.5	0.6	2.2	3.9	7.5	11.1	16.2	17.7	16.1	11.0	6.5	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.5	6.4	7.7	11.5	15.0	19.8	20.7	19.2	15.2	9.9	5.7
		MCDB	3.9	3.7	7.2	9.3	13.0	16.9	21.6	22.3	19.8	15.9	10.5	5.8
	2%	WB	2.6	2.6	4.6	6.0	9.7	13.5	18.5	19.7	18.0	13.6	8.6	4.7
		MCDB	2.9	2.8	5.5	7.4	11.0	15.0	20.2	21.0	18.9	14.5	9.0	5.0
	5%	WB	1.8	1.8	3.2	5.0	8.8	12.4	17.3	18.9	17.1	12.3	7.5	4.0
		MCDB	2.1	2.1	3.9	6.1	9.9	13.8	18.9	20.1	18.1	13.2	8.0	4.4
	10%	WB	0.5	0.7	2.3	4.2	7.8	11.4	16.2	17.9	16.3	11.3	6.6	3.1
		MCDB	1.0	1.1	2.9	5.2	8.7	12.6	17.7	19.3	17.3	12.2	7.2	3.7
Mean Daily Temperature Range	5% DB	MDBR	5.3	4.9	4.8	4.3	4.9	5.6	5.3	4.5	3.8	3.3	3.6	4.3
		MCDBR	4.7	3.9	4.9	5.8	7.0	7.7	6.4	5.3	4.4	3.8	3.4	3.9
		MCWBR	4.7	3.8	3.7	4.0	4.9	5.5	4.2	3.4	3.2	3.2	3.2	3.8
	5% WB	MCDBR	4.9	3.7	4.4	5.4	6.5	7.0	6.0	5.0	4.3	3.7	3.3	3.9
		MCWBR	5.0	3.7	3.5	4.0	5.2	5.6	4.3	3.5	3.3	3.3	3.3	3.9
Clear-Sky Solar Irradiance	taub	0.286	0.286	0.310	0.342	0.371	0.381	0.390	0.373	0.364	0.337	0.304	0.279	
	taud	2.480	2.468	2.415	2.368	2.382	2.382	2.399	2.445	2.458	2.503	2.522	2.524	
	Ebn at Noon	812	893	916	912	893	883	868	871	846	818	782	782	
	Edh at Noon	64	79	97	113	116	117	113	104	94	77	62	55	
All-Sky Solar Radiation	RadAvg	0.99	1.86	3.32	4.64	5.57	6.02	6.04	5.16	3.73	2.19	1.07	0.72	
	RadStd	0.11	0.23	0.41	0.51	0.42	0.40	0.34	0.35	0.24	0.26	0.16	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page